

Usama Puward

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in usama-puward | 🌱 Usamafuward

Profile

AI/SE Engineer and Software Developer with a Computer Science background, experience in Software Engineering and Artificial Intelligence. Dedicated to developing innovative, scalable solutions, tackling complex challenges, and driving real-world impact through collaboration and continuous learning.

[Explore more of my works] [🔗](#).

Education

BSc in Computer Science, University of Colombo

May 2022 – Aug 2025

- **Coursework:** Software Engineering, Data Structure and Algorithms, Machine Learning and Neural Computing, Middleware Architecture, SQA, DBMS

Professional Experience

AI/SE Engineer, Kainovation Technologies – Colombo, Sri Lanka

Nov 2024 – Present

- Designed and implemented the **Support AI Assistant**, a full-stack conversational AI with chat and voice support, using **Next.js** (frontend) and **FastAPI** (backend) integrated with **OpenAI Realtime API**, **Gemini**, and **vector stores**, delivering real-time, context-aware support for customers and employees
- Built an **Intelligent Document OCR Solution** with **FastHTML** (frontend) and **FastAPI** (backend), integrating **Gemini** and **OpenAI OCR models** to extract data from handwritten and printed documents with high accuracy
- Developed **Voice AI solutions** (AI Recruitment Assistant, AI Insurance Agent, AI Receptionist) with **Next.js** (frontend) and **FastAPI + Hume AI** (backend), enabling automated interviews, policy explanation, and real-time voice-enabled scheduling with calendar sync
- Created the **SAP Invoice Automation** frontend in **FastHTML**, streamlining invoice upload and formatting for SAP-ready outputs, improving accuracy and reducing manual data entry
- Developed the **InsurePulse landing website** using **React** and **Next.js**, delivering a responsive, SEO-optimized showcase for the platform's reporting and analytics features
- Independently developed the **official Kainovation Technologies website** in **Next.js**, achieving optimized performance, SEO visibility, and a fully responsive user experience
- Contributed to the **InsurePulse** core architecture with **Next.js**, **NestJS**, and **Prisma ORM**, integrating real-time analytics dashboards (claims, underwriting) with multi-database ETL pipelines (**Oracle**, **Postgres**)
- Collaborated on **InsurePal**, an AI conversational BI assistant within InsurePulse, built with **Next.js**, **FastAPI**, and **Claude**, enabling natural language to SQL translation and streaming query outputs into interactive **ApexCharts** via SSE

Contract Software Developer, Edus Lanka (PVT) LTD – Jaffna, Sri Lanka

Nov 2024 – Jan 2025

- Contributed to the development of **Mediman – Doctor Patient Clinic Online Appointment System**, a **full-stack MERN project**
- Built the frontend using **React** and **Next.js**, delivering a responsive and user-friendly interface for appointment booking, patient management, and real-time doctor-patient communication
- Implemented **payment gateway integration** for secure online transactions, ensuring smooth and reliable payment flows
- Developed the **admin dashboard** for doctors and staff, enabling efficient management of appointments, patients, and system configurations

Projects

Eats Robers – Food Ordering Web App

github.com/Usamafuward [↗](#)

- Designed and developed a full-stack application for discovering meals, placing orders, and managing payments
- Implemented RESTful APIs and MongoDB integration for seamless user experience
- Tools Used: React, Node.js, Express.js, MongoDB, Mongoose

Travel Point – Social Travel Platform

github.com/Usamafuward [↗](#)

- Built a social platform for travelers to share itineraries, experiences, and book accommodations
- Integrated FastAPI backend with React Native frontend for seamless web and mobile experience
- Implemented authentication, PostgreSQL database, and responsive UI using Tailwind CSS
- Tools Used: React, React Native, Node.js, FastAPI, PostgreSQL, Tailwind CSS

Tartuca – Restaurant and Delivery App

User [↗](#), Admin [↗](#), Backend [↗](#)

- Developed a full-stack restaurant system with a customer app, admin portal, and RESTful API backend.
- Built an asynchronous FastAPI backend integrated with PostgreSQL and Alembic database migrations.
- Integrated Auth0 for user authentication and Google Maps API for delivery location pinning and autocomplete.
- Tools Used: React, FastAPI, PostgreSQL, SQLAlchemy, Alembic, Auth0, Google Maps API, Vite

Technologies

Languages: Python, JavaScript, TypeScript, C++, SQL **AI/ML:** TensorFlow, OpenAI API, Gemini, Hume AI, Pandas, NumPy, OpenCV, LangChain, Vector Databases **Frontend:** React, Next.js, React Native, FastHTML, Tailwind CSS **Backend:** FastAPI, Node.js, Express, Django, MongoDB, PostgreSQL **Cloud & DevOps:** Microsoft Azure, Git, Docker, RESTful APIs

Certifications

Meta React Specialization [↗](#)

Meta

Machine Learning Specialization [↗](#)

DeepLearning.AI & Stanford

Introduction to Front-End Development [↗](#), Mobile Development [↗](#)

Meta

Introduction to TensorFlow [↗](#), CNNs in TensorFlow [↗](#)

DeepLearning.AI

Artificial Intelligence on Microsoft Azure [↗](#)

Microsoft

Azure Fundamentals Training [↗](#)

Styava

Developing Back-End Apps with Node.js and Express [↗](#)

IBM

Mastering Multi-Agent Development with AutoGen [↗](#)

Packt

DevOps Essentials and Version Control with Git [↗](#)

Edureka

30 Days MasterClass in Artificial Intelligence [↗](#)

NoviTech R&D

Introduction to Generative AI [↗](#), Language Models [↗](#), Responsible AI [↗](#)

Google Cloud

References

Mr. M.R. Sajeer Ahamed – MSc in Computer Science, Nottingham Trent University
Software Development Engineer, Amazon, London, UK

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